

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Forged Steel Yoke/Knuckle-type Component (PRCR002)

Commodity: FORGING · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 2 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥391.31	28.3%	39.0%	10.5%	7.4%

Model Confidence: **Low** · 2 traced operations · 7 data points auditable

Engine Warnings (1)

rawMaterial.materialId: Material rate confidence: Medium

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 1037.1 cm³; mass for the selected material family 8.141 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: forging

Alloy / material: 1020 / S20C · Net weight: 8.141 kg (measured)

Material utilisation: 68% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 2

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥110.65	28.3%	
2. Process (Machine)	¥152.67	39.0%	
3. Direct Labour	¥40.92	10.5%	
4. Tooling (amortised)	¥9.81	2.5%	
5. Packaging	¥3.53	0.9%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
6. Logistics	¥7.06	1.8%	
Factory Cost	¥324.64	83.0%	
7. Overhead (SG&A)	¥37.69	9.6%	
Subtotal	¥362.32	92.6%	
8. Supplier Margin	¥28.99	7.4%	
TOTAL SHOULD-COST	¥391.31	100.0%	

§2 — Commercial Parameters

Overhead Rate	12.0%	Supplier Margin	8.0%
Packaging / part	¥3.53	Logistics / part	¥7.06
Total Tooling Cost	¥1962407.24	Amortisation Volume	200,000 parts

Geometric DFM / DFA - measured from the CAD model

2 issue(s) across 24 instance(s), from 128 measured feature(s) and 4 rule(s). Every issue names the B-rep faces that produced it and the published source of its threshold. No finding on this part has a modelled cost path; each says why.

[MAJOR] Fillet too sharp for hot metal flow (18 instances)

Worst: Fillet at face 177 is R1.50 mm against a 3 mm minimum for hot forging.
 measured radiusMm 1.5-2mm threshold < 3mm
 Faces: 34, 64, 69, 117, 119, 126, 169, 174, 177, 227, 279, 295, 296, 305, 308, 311, 313, 362
 Fix: Open the fillet to at least R3 mm. Hot metal will not turn a sharp corner: it folds instead, producing laps that only show up at magnetic-particle inspection, and the sharp die corner is where the die itself cracks first.
 Source: ASM Handbook Vol. 14A, Metalworking: Bulk Forming — fillet and corner radii - Minimum fillet ra
 Mirrors the 3 mm threshold in analyseForgingDFM. Cold forming is exempt: it flows metal at room temperature into tighter radii, so the rule does not run for that route.

[MAJOR] Undercut — cannot release from an impression die (6 instances)

Worst: Face 172 sits at 120.0° to the draw (past 90°) and cannot lift out of an impression die.
 measured angleToDrawDeg 94.57-120° threshold > 90°
 Faces: 10, 11, 35, 172, 178, 194
 Fix: Move the parting line, or accept the feature as a machined operation after forging and carry that cost. Unlike casting or moulding, a forging die cannot buy its way out with a slide.
 Source: ASM Handbook Vol. 14A, Metalworking: Bulk Forming — die design and parting
 A forging die has no slides or cores: the part must lift straight out of the impression. A face whose normal opposes the draw cannot be forged on that parting and has to be machined afterwards, or the parting has to move.

DFA - handling & insertion (Boothroyd method, geometric half)

Handling 1.63s + insertion 4.3s = 5.93s, 1.98x the 3s ideal part.

+0.5s Asymmetric envelope — the part must be oriented before insertion (bounding box 277×223×182 mm, no two dimensions within 5%)

+2.8s Fasteners approach from more than one direction — the part must be re-oriented or the fixture indexed during assembly (5 distinct hole axes across 26 holes)

Theoretical minimum part count, and therefore the Boothroyd design-efficiency index, requires the full assembly — this analysis covers one part, so no efficiency ratio is given.

Nesting and tangling risk is not assessed: it depends on how parts are presented in bulk, which the geometry alone does not determine.

Rotational symmetry is approximated from the bounding box, not from an inertia-tensor shape comparison.

What was NOT checked

No material family was confirmed, so any check whose threshold depends on the alloy or resin did not run.

Grain-flow alignment was not assessed. Whether the forged flow lines follow the principal load path is the single largest reason to forge a part rather than cast or machine it, and it depends on the die design and preform, not on the finished shape.

Parting-line position was assumed perpendicular to the draw. A different parting changes which faces are undercuts and where the flash line falls; if the flash line crosses a sealing or highly-loaded face that is a defect the geometry cannot reveal.

Machining stock allowance on forged surfaces was not checked — it is a process decision.

Rib height-to-thickness was not computed: it needs rib identification, and the kernel does not yet separate a rib from any other thin planar region.

Billet and preform design, and therefore realistic flash loss, were not evaluated.

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-steel1020	ID	
Grade / Specification	1020 / S20C		UK steel billet, Jun 2026. Index-anchored 2026-07 (see FORGING PRICING BASIS). Regional adj. ×0.880 (millSteel)
Region	China		
Net Finished Weight	8.1410 kg	kg	Weight in finished part
Gross Weight (stock)	12.0525 kg	kg	Net ÷ utilisation ratio
Scrap / Runner Weight	3.9115 kg	kg	Gross - Net
Material Utilisation	67.5%	%	Benchmark: forging 75–90 % (flash + bar drop)
Material Price	¥6.53	CNY /kg	UK steel billet, Jun 2026. Index-anchored 2026-07 (see FORGING PRICING BASIS). Regional adj. ×0.880 (millSteel)
Scrap Recovery Price	¥1.51	CNY /kg	
Gross Material Cost	¥78.71	CNY	Gross × price/kg
Scrap Credit	-¥5.92	CNY	Scrap × recovery price
Consumables (core / wax / shell)	¥37.86	CNY	Per-part recurring
NET RAW MATERIAL COST	¥110.65	CNY	Gross - scrap credit + consumables
Data Confidence	Medium		
Effective Date	2026-07		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Forging	1600T Mechanical Forge Press	¥416.21	1.33	1	80.0%	¥11.56
CNC Machining — 29 features (1×Ø3.1×3, 1×Ø6.3×4, 1×Ø10.0×23, 2×Ø14.0×8, 2×Ø14.0×40, 2×Ø14.0×59, 1×Ø16.0×3, 1×Ø24.0×18, 1×Ø37.0×45,	CNC VMC 3-axis	¥254.30	28.30	1	85.0%	¥141.10

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
1×Ø39.0×45, 2×Ø40.0×3, 2×Ø40.0×7, 2×Ø42.5×2, 1×Ø45.0×3, 1×Ø46.0×6, 1× face 10594mm², 1× face 4683mm², 1× face 3158mm², 1× face 2940mm², 1× face 2381mm², 1× face 2212mm², 1× face 1046mm², 1× face 910mm²) [geometry-measured]						
TOTAL						¥152.67

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Forging	Skilled Machinist	¥71.49	2	1.33	92.0 %	¥3.45	¥15.02
CNC Machining — 29 features (1×Ø3.1×3, 1×Ø6.3×4, 1×Ø10.0×23, 2×Ø14.0×8, 2×Ø14.0×40, 2×Ø14.0×59, 1×Ø16.0×3, 1×Ø24.0×18, 1×Ø37.0×45, 1×Ø39.0×45, 2×Ø40.0×3, 2×Ø40.0×7, 2×Ø42.5×2, 1×Ø45.0×3, 1×Ø46.0×6, 1× face 10594mm², 1× face 4683mm², 1× face 3158mm², 1× face 2940mm², 1× face 2381mm², 1× face 2212mm², 1× face 1046mm², 1× face 910mm²) [geometry-measured]	Skilled Machinist	¥71.49	1	28.30	90.0 %	¥37.47	¥178.57
TOTAL						¥40.92	¥193.58

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Boss	Ø20.0 × 4	1	Turning (external Ø)	0.38	—	No
Boss	Ø24.0 × 25	1	Turning (external Ø)	0.79	—	No
Boss	Ø35.0 × 47	1	Turning (external Ø)	1.24	—	No
Boss	Ø55.0 × 32	1	Turning (external Ø)	0.93	—	No
Boss	Ø57.0 × 7	1	Turning (external Ø)	0.44	—	No
Boss	Ø77.0 × 2	1	Turning (external Ø)	0.34	—	No
Boss	Ø114.0 × 7	1	Turning (external Ø)	0.44	—	No
Hole / bore	Ø3.1 × 3 blind	1	Drilling (blind)	0.19	¥0.81	Yes
Hole / bore	Ø6.3 × 4 blind	1	Drilling (blind)	0.32	¥1.37	Yes
Hole / bore	Ø10.0 × 23 blind	1	Drilling (blind)	0.70	¥2.97	Yes
Hole / bore	Ø14.0 × 8 blind	2	Drill + bore (blind)	1.64	¥6.95	Yes

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Hole / bore	Ø14.0 × 40 blind	2	Drill + bore (blind)	4.16	¥17.63	Yes
Hole / bore	Ø14.0 × 59 blind	2	Drill + bore (blind)	5.72	¥24.24	Yes
Hole / bore	Ø16.0 × 3 blind	1	Drill + bore (blind)	0.63	¥2.66	Yes
Hole / bore	Ø24.0 × 18 blind	1	Drill + bore (blind)	1.20	¥5.09	Yes
Hole / bore	Ø37.0 × 45 blind	1	Drill + bore (blind)	2.85	¥12.08	Yes
Hole / bore	Ø39.0 × 45 blind	1	Drill + bore (blind)	2.85	¥12.08	Yes
Hole / bore	Ø40.0 × 3 blind	2	Drill + bore (blind)	0.72	¥3.05	Yes
Hole / bore	Ø40.0 × 7 blind	2	Drill + bore (blind)	0.72	¥3.05	Yes
Hole / bore	Ø42.5 × 2 blind	2	Drill + bore (blind)	0.74	¥3.14	Yes
Hole / bore	Ø45.0 × 3 blind	1	Drill + bore (blind)	0.38	¥1.61	Yes
Hole / bore	Ø46.0 × 6 blind	1	Drill + bore (blind)	0.38	¥1.63	Yes
Face	10594 mm²	1	Face milling (facing)	1.52	¥6.46	Yes
Face	4683 mm²	1	Face milling (facing)	0.79	¥3.33	Yes
Face	3158 mm²	1	Face milling (facing)	0.59	¥2.52	Yes
Face	2940 mm²	1	Face milling (facing)	0.57	¥2.41	Yes
Face	2381 mm²	1	Face milling (facing)	0.50	¥2.11	Yes
Face	2212 mm²	1	Face milling (facing)	0.48	¥2.02	Yes
Face	1046 mm²	1	Face milling (facing)	0.33	¥1.40	Yes
Face	910 mm²	1	Face milling (facing)	0.31	¥1.33	Yes
TOTAL costed		29		28.30	¥119.94	Yes

7 detected feature(s) NOT costed

7 machinable feature instance(s) (4.6 min) were detected in the geometry but are NOT included in this cost — on a near-net part they may be as-cast. If they are machined, tick them in the Machined-Features panel and re-cost; this could add up to ¥19.36.

\$5 — Machine Rate Buildup

CNC VMC 3-axis [mach-vmc3] · Confidence: Low · UK Tier-2 3-axis VMC (HAAS VF/VM class). Target £55/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥273755.66	¥85.55	4,000 hr/yr available
Maintenance	¥149321.27	¥46.66	
Energy	¥77119.81	¥24.10	
Floor Space	¥59728.51	¥18.67	
Indirect Support	¥149321.27	¥46.66	
Finance Cost	¥104524.89	¥32.66	
TOTAL — CNC VMC 3-axis	¥813771.39	¥254.30	Effective hours: 3200/yr

1600T Mechanical Forge Press [forge-press-1600t] · Confidence: Low · UK forge shop benchmark (mid-size crank press), Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @78% util	Notes
Depreciation	¥447963.80	¥164.09	3,500 hr/yr available
Maintenance	¥209049.77	¥76.58	
Energy	¥234113.71	¥85.76	
Floor Space	¥99547.51	¥36.46	
Indirect Support	¥89592.76	¥32.82	
Finance Cost	¥55995.48	¥20.51	
TOTAL — 1600T Mechanical Forge Press	¥1136263.03	¥416.21	Effective hours: 2730/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

\$6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
material.pricePerKg	0.7216	£/kg	UK steel billet, Jun 2026. Index-anchored 2026-07 (see FORGING PRICING BASIS). Regional adj.	mat-steel1020	Medium

Field	Value	Unit	Source / Reference	Rate ID	Conf.
			×0.880 (millSteel)		
material.scrapRecoveryPricePerKg	0.1672	£/kg	UK steel billet, Jun 2026. Index-anchored 2026-07 (see FORGING PRICING BASIS). Regional adj. ×0.880 (millSteel)	mat-steel1020	Medium
rawMaterial.consumablesCostPerPart	4.1839	£	Per-part consumable (core/wax/shell)	mat-steel1020	Medium
Forging.machineRatePerHr	45.9916	£/hr	UK forge shop benchmark (mid-size crank press), Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	forge-press-1600t	Low
Forging.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
CNC Machining — 29 features (1×Ø3.1×3, 1×Ø6.3×4, 1×Ø10.0×23, 2×Ø14.0×8, 2×Ø14.0×40, 2×Ø14.0×59, 1×Ø16.0×3, 1×Ø24.0×18, 1×Ø37.0×45, 1×Ø39.0×45, 2×Ø40.0×3, 2×Ø40.0×7, 2×Ø42.5×2, 1×Ø45.0×3, 1×Ø46.0×6, 1× face 10594mm², 1× face 4683mm², 1× face 3158mm², 1× face 2940mm², 1× face 2381mm², 1× face 2212mm², 1× face 1046mm², 1× face 910mm²) [geometry-measured].machineRatePerHr	28.1005	£/hr	UK Tier-2 3-axis VMC (HAAS VF/VM class). Target £55/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-vmc3	Low
CNC Machining — 29 features (1×Ø3.1×3, 1×Ø6.3×4, 1×Ø10.0×23, 2×Ø14.0×8, 2×Ø14.0×40, 2×Ø14.0×59, 1×Ø16.0×3, 1×Ø24.0×18, 1×Ø37.0×45, 1×Ø39.0×45, 2×Ø40.0×3, 2×Ø40.0×7, 2×Ø42.5×2, 1×Ø45.0×3, 1×Ø46.0×6, 1× face 10594mm², 1× face 4683mm², 1× face 3158mm², 1× face 2940mm², 1× face 2381mm², 1× face 2212mm², 1× face 1046mm², 1× face 910mm²) [geometry-measured].labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥391.31 ± 17.1%	¥327.06	¥387.78	¥460.63	moderate	13.4%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 4 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread; some detected features not costed (see §4C).

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.

Import duty and international freight (regional table is Ex-Works).

Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
CNC Machining — 29 features (1×Ø3.1×3, 1×Ø6.3×4, 1×Ø10.0×23, 2×Ø14.0×8, 2×Ø14.0×40, 2×Ø14.0×59, 1×Ø16.0×3, 1×Ø24.0×18, 1×Ø37.0×45, 1×Ø39.0×45, 2×Ø40.0×3, 2×Ø40.0×7, 2×Ø42.5×2, 1×Ø45.0×3, 1×Ø46.0×6, 1× face 10594mm², 1× face 4683mm², 1× face 3158mm², 1× face 2940mm², 1× face 2381mm², 1× face 2212mm², 1× face 1046mm², 1× face 910mm²) [geometry-measured]: Cycle Time	0.47hr	-5.5%	+5.5%	¥43.20
CNC Machining — 29 features (1×Ø3.1×3, 1×Ø6.3×4, 1×Ø10.0×23, 2×Ø14.0×8, 2×Ø14.0×40, 2×Ø14.0×59, 1×Ø16.0×3, 1×Ø24.0×18, 1×Ø37.0×45, 1×Ø39.0×45, 2×Ø40.0×3, 2×Ø40.0×7, 2×Ø42.5×2, 1×Ø45.0×3, 1×Ø46.0×6, 1× face 10594mm², 1× face 4683mm², 1× face 3158mm², 1× face 2940mm², 1× face 2381mm², 1× face 2212mm², 1× face 1046mm², 1× face 910mm²) [geometry-measured]: Machine Rate	¥254.30/hr	-4.4%	+4.4%	¥34.14
Material: 1020 / S20C	¥6.53/kg	-2.4%	+2.4%	¥19.04
Material utilisation	0.68	+2.1%	-1.7%	¥14.78
Forging: Labour Rate	¥71.49/hr	-1.3%	+1.3%	¥9.90
CNC Machining — 29 features (1×Ø3.1×3, 1×Ø6.3×4, 1×Ø10.0×23, 2×Ø14.0×8, 2×Ø14.0×40, 2×Ø14.0×59, 1×Ø16.0×3, 1×Ø24.0×18, 1×Ø37.0×45, 1×Ø39.0×45, 2×Ø40.0×3, 2×Ø40.0×7, 2×Ø42.5×2, 1×Ø45.0×3, 1×Ø46.0×6, 1× face 10594mm², 1× face 4683mm², 1× face 3158mm², 1× face 2940mm², 1× face 2381mm², 1× face 2212mm², 1× face 1046mm², 1× face 910mm²) [geometry-measured]: Labour Rate	¥71.49/hr	-1.3%	+1.3%	¥9.90
Overhead %	12%	-1%	+1%	¥8.14

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Supplier Margin %	8%	-0.7%	+0.7%	¥5.80

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥125.74	¥277.57	¥147.31	¥17.84	¥50.25	¥618.71	¥4.87	¥657.60	+68%
Germany (EUR)	¥129.51	¥291.45	¥238.07	¥18.73	¥65.90	¥743.67	¥5.60	¥789.36	+102%
France (EUR)	¥128.25	¥255.37	¥171.12	¥16.41	¥53.01	¥624.16	¥5.60	¥664.33	+70%
Spain (EUR)	¥125.74	¥244.26	¥107.88	¥15.70	¥41.45	¥535.03	¥5.60	¥570.93	+46%
Poland (PLN)	¥121.97	¥199.85	¥66.96	¥12.84	¥30.18	¥431.80	¥5.84	¥462.57	+18%
Turkey (TRY)	¥113.17	¥166.54	¥37.20	¥10.70	¥22.59	¥350.20	¥6.33	¥377.19	-4%
China (CNY)	¥110.65	¥152.67	¥40.92	¥9.81	¥37.69	¥351.73	¥7.06	¥391.31	Base
India (INR)	¥113.17	¥144.34	¥26.04	¥9.28	¥18.64	¥311.46	¥7.30	¥336.88	-14%
Mexico (MXN)	¥119.45	¥166.54	¥43.15	¥10.70	¥23.43	¥363.28	¥6.57	¥390.86	-0%
United States (USD)	¥125.74	¥235.94	¥193.43	¥15.16	¥47.89	¥618.16	¥6.09	¥658.04	+68%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

40.84 kgCO₂e · 5.02 /kg · cradle-to-gate

Material	18.33 kgCO ₂ e	Material factor	2.10 kg/kg (Steel)
Process	21.21 kgCO ₂ e	Grid intensity	0.550 kg/kWh · 38.57 kWh
Logistics	1.30 kgCO ₂ e	Total	40.84 kgCO ₂ e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[WARNING] Machining / process cost above benchmark

Finding	Process cost at 39.0% of total is above the benchmark range of 10–20%. Cycle time or machine rate is elevated.
Impact	High · up to 10% potential saving
Benchmark	Process %: yours 39.0% vs industry 10–20%
Action 1	Reduce cycle time via feeds/speeds optimisation and toolpath strategy review
Action 2	Combine multiple operations into one setup to eliminate re-fixture time

[WARNING] Low material utilisation — high scrap rate

Finding	Material utilisation at 68% is below the forging benchmark of 75%. You are paying for 48% more raw material than ends up in the part.
Impact	Medium · up to 1% potential saving
Action 1	Redesign blank/preform geometry to minimise offcuts
Action 2	Optimise nesting/layout for sheet or profile stock

[INFO] Material cost is below benchmark range

Finding	Material at 28.3% is below the typical 38–62% range. Conversion cost is the dominant driver.
Impact	Low
Benchmark	Material %: yours 28.3% vs industry 38–62%
Action 1	Focus optimisation efforts on process and labour efficiency

§12 — Cost-Reduction Opportunities (ranked by saving)

7 opportunities · combined ¥52.04/part (~13%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Material	Improve Material Utilisation via Near-Net-Shape	Current utilisation at 68%. Near-net-shape process or improved nesting targets 80–90%.	¥34.15	8.7%	Medium Term	Medium
2	Material	Consumables Rationalisation (Cores, Patterns, Shell)	Per-part consumables are 34% of the material line. Core count, pattern life and shell recipe are all design-controllable.	¥23.48	6.0%	Medium Term	Medium
3	Material	Scrap Revenue	32% of the bought material leaves as chips,	¥11.22	2.9%	Quick Win	Low

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
		Recovery at Index Prices	offcuts or skeleton. That scrap has an index price — the credit belongs in the piece price, not in the supplier's margin.				
4	Process & Automation	Attack the Bottleneck: "CNC Machining — 29 features (1×Ø3.1×3, 1×Ø6.3×4, 1×Ø10.0×23, 2×Ø14.0×8, 2×Ø14.0×40, 2×Ø14.0×59, 1×Ø16.0×3, 1×Ø24.0×18, 1×Ø37.0×45, 1×Ø39.0×45, 2×Ø40.0×3, 2×Ø40.0×7, 2×Ø42.5×2, 1×Ø45.0×3, 1×Ø46.0×6, 1× face 10594mm², 1× face 4683mm², 1× face 3158mm², 1× face 2940mm², 1× face 2381mm², 1× face 2212mm², 1× face 1046mm², 1× face 910mm²) [geometry-measured]"	One operation — CNC Machining — 29 features (1×Ø3.1×3, 1×Ø6.3×4, 1×Ø10.0×23, 2×Ø14.0×8, 2×Ø14.0×40, 2×Ø14.0×59, 1×Ø16.0×3, 1×Ø24.0×18, 1×Ø37.0×45, 1×Ø39.0×45, 2×Ø40.0×3, 2×Ø40.0×7, 2×Ø42.5×2, 1×Ø45.0×3, 1×Ø46.0×6, 1× face 10594mm², 1× face 4683mm², 1× face 3158mm², 1× face 2940mm², 1× face 2381mm², 1× face 2212mm², 1× face 1046mm², 1× face 910mm²) [geometry-measured] — carries 92% of the conversion cost. Improving anything else first is working on the wrong problem.	¥31.30	8.0%	Medium Term	Medium
5	Process & Automation	Line-balance or automate the multi-manned station first; check whether the second operator exists only to load/unload	High manning (2.0 operators on one operation) — At least one operation runs with 2.0 operators. Multi-manned stations are the strongest automation candidates on any part.	¥23.48	6.0%	Medium Term	Medium
6	Commercial & Sourcing	Annual Volume Re-Commitment for Better Pricing	Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	¥11.74	3.0%	Quick Win	Low
7	Sustainability & Energy	Energy Productivity Programme	This process is energy-intensive (melt, cure, oven or heat-treat content). Energy sits inside the machine rate and overhead — and it is the fastest-moving input cost of the decade.	¥11.74	3.0%	Medium Term	Low

§13 — Opportunity by Category

4 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Material	3	Improve Material Utilisation via Near-Net-Shape	¥34.15	¥68.85
Process & Automation	2	Attack the Bottleneck: "CNC Machining — 29 features (1×Ø3.1×3, 1×Ø6.3×4, 1×Ø10.0×23, 2×Ø14.0×8, 2×Ø14.0×40, 2×Ø14.0×59, 1×Ø16.0×3, 1×Ø24.0×18, 1×Ø37.0×45, 1×Ø39.0×45, 2×Ø40.0×3, 2×Ø40.0×7, 2×Ø42.5×2, 1×Ø45.0×3, 1×Ø46.0×6, 1× face 10594mm², 1× face 4683mm², 1× face 3158mm², 1× face 2940mm², 1× face 2381mm², 1× face 2212mm², 1× face 1046mm², 1× face 910mm²) [geometry-measured]"	¥31.30	¥54.78
Commercial & Sourcing	1	Annual Volume Re-Commitment for Better Pricing	¥11.74	¥11.74
Sustainability & Energy	1	Energy Productivity Programme	¥11.74	¥11.74

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Scrap Revenue Recovery at Index Prices
- Annual Volume Re-Commitment for Better Pricing